

ECON 712: Applied Econometrics

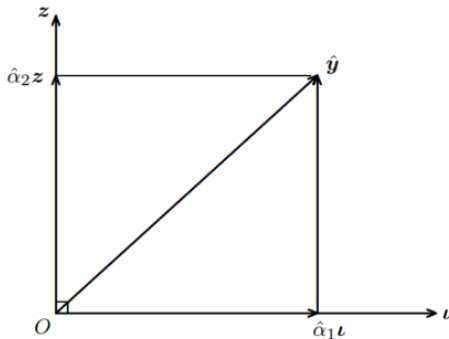
Lecture 9, 10: Linear Regression and the CEF

FWL Theorem– Using Vector Algebra

CEF and Linear Regression

Linear Transformations

- Subtracting a constant $\iota\bar{x}$ from regressors when the constant term is included



Orthogonal regressors may be omitted

Linear Transformations

- For a general regression where regressors are partitioned:
 $\mathbf{X} = [\mathbf{X}_1 \mathbf{X}_2]$, $\mathbf{X}_1$ is an $n \times k_1$, $\mathbf{X}_2$ is an $n \times k_2$; β_1 is $k_1 \times 1$, β_2 is $k_2 \times 1$,
and $k_1 + k_2 = k$

$$\mathbf{y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + u$$

the OLS/MM estimator has the following properties:

$$\mathbf{y} = \mathbf{X}_1\hat{\beta}_1 + \mathbf{X}_2\hat{\beta}_2 + \hat{u}$$

$$\mathbf{y} = \mathbf{X}_1\hat{\alpha}_1 + \mathbf{Z}\hat{\beta}_2 + \hat{u}$$

where $\mathbf{Z} = \mathbf{X}_2 - \mathbf{X}_1 \mathbf{A}$ and $\mathbf{A}$ is a $k_1 \times k_1$ non-singular matrix

Linear Transformations

- In addition, if the transformation results in:

$$\langle \mathbf{X}_1, \mathbf{Z} \rangle = \mathbf{X}_1^T \mathbf{Z} = \mathbf{0}$$

then,

$$\mathbf{y} = \mathbf{Z}\hat{\beta}_2 + \hat{\epsilon}$$

- This is the basis of the FWL theorem we studied in ECON 312:

- ▶ Let $\mathbf{Z} = \mathbf{X}_2 - \mathbf{P}_{\mathbf{X}_1} \mathbf{X}_2 = \mathbf{M}_{\mathbf{X}_1} \mathbf{X}_2$, then

$$\mathbf{P}_Z \mathbf{y} = \mathbf{Z}\hat{\beta}_2$$

- Note that residuals of this regression are not the same as the residuals of the original regression

FWL– with vector algebra

- Suppose that the OLS/MM estimator results in:

$$\mathbf{y} = \mathbf{X}_1 \hat{\beta}_1 + \mathbf{X}_2 \hat{\beta}_2 + \hat{\mathbf{u}}$$

and define $\mathbf{Z} = \mathbf{M}_{\mathbf{X}_1} \mathbf{X}_2$, $\tilde{\mathbf{y}} = \mathbf{M}_{\mathbf{X}_1} \mathbf{y}$, then:

$$\mathbf{P}_Z \tilde{\mathbf{y}} = \mathbf{Z} \hat{\beta}_2 \quad , \text{ and}$$

$$\mathbf{M}_Z \tilde{\mathbf{y}} = \hat{\mathbf{u}}$$

- This yields the following equation, where $\hat{\beta}_2$ denotes the OLS/MM estimator:

$$\tilde{\mathbf{y}} = \mathbf{Z} \hat{\beta}_2 + \hat{\mathbf{u}}$$

Goodness of Fit

- Recall that, by Pythagoras' Theorem:

$$TSS = \|\mathbf{y}\|^2 = \|\mathbf{P}_X \mathbf{y}\|^2 + \|\mathbf{M}_X \mathbf{y}\|^2 = ESS + SSR$$

- We use this result to obtain two goodness of fit measures:

$$R_u^2 = \frac{\|\mathbf{P}_X \mathbf{y}\|^2}{\|\mathbf{y}\|^2}$$

$$R_c^2 = \frac{\|\mathbf{P}_X \mathbf{M}_\iota \mathbf{y}\|^2}{\|\mathbf{M}_\iota \mathbf{y}\|^2}$$

- Note that the R^2 -centered is preferred, but it can only be interpreted when the constant term is included in the regression

FWL Theorem– Using Vector Algebra

CEF and Linear Regression

Why Linear Regression?

- Suppose that the population CEF is linear and has the following functional form:

$$\mathbb{E}[y_i|\mathbf{X}_i] = \mathbf{X}_i^T \beta^* \text{ (MHE notation)}$$

- ▶ by the CEF Decomposition Theorem:

$$\mathbb{E}[\mathbf{X}_i(y_i - \mathbb{E}[y_i|\mathbf{X}_i])] = \mathbf{0}$$

- ▶ Substituting for the linear CEF:

$$\mathbb{E}[\mathbf{X}_i(y_i - \mathbf{X}_i^T \beta^*)] = \mathbf{0}$$

- ▶ Solving for $\beta^* = \mathbb{E}[\mathbf{X}_i \mathbf{X}_i^T]^{-1} \mathbb{E}[\mathbf{X}_i y_i] = \beta^{OLS}$

- Interpretation: OLS solves the same problem as the linear CEF

Why Linear Regression?

- What if the CEF is **not linear**?

1. By definition, β^{OLS} provides the best linear prediction of y_i :

$$\beta^{OLS} = \underset{b}{\operatorname{argmin}} \mathbb{E}[(y_i - \mathbf{X}_i^T \mathbf{b})^2]$$

2. β^{OLS} is the best linear approximation to the CEF:

$$\beta = \underset{b}{\operatorname{argmin}} \mathbb{E}[(\mathbb{E}[y_i | \mathbf{X}_i] - \mathbf{X}_i^T \mathbf{b})^2]$$

- Proof: by FOC

$$\mathbb{E}[\mathbf{X}_i(\mathbb{E}[y_i | \mathbf{X}_i] - \mathbf{X}_i^T \beta)] = \mathbf{0} \Rightarrow$$

$$\beta = \mathbb{E}[\mathbf{X}_i \mathbf{X}_i^T]^{-1} \mathbb{E}[\mathbf{X}_i y_i] = \beta^{OLS}$$

where the last line uses the CEF decomposition property.

Linear Regression and CEF



Finite Sample Properties of OLS

- Under strict exogeneity, $\mathbb{E}[\mathbf{u}|\mathbf{X}] = \mathbf{0}$, OLS/MM is unbiased:

$$\hat{\beta}^{ols} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$\hat{\beta}^{ols} = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{u}$$

$$\mathbb{E}[\hat{\beta}^{ols}|\mathbf{X}] = \beta \underbrace{=}_{\text{by LIE}} \mathbb{E}[\hat{\beta}^{ols}]$$

- Pre-determinedness, $\mathbb{E}[u_i|x_{1i}, \cdot, x_{ki}] = 0$, is not enough
- The linear CEF has to be correctly specified